
High-Dimensional Sensitivity Analysis for Genomic Studies: An Adversarial Framework for Learning Worst-Case Latent Confounders

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Abstract

High-dimensional genomics studies are frequently confounded by unmeasured biological processes that obscure disease-specific signals. While existing workflows can estimate these latent confounders, they fail to quantify how robust a discovery is to varying levels of hypothetical confounding. We introduce sensGAN, a deep-learning adversarial framework that systematically explores the confounding spectrum by learning “worst-case” latent variables that nullify the most gene associations under novel predictive-gain constraints. By identifying the minimum confounding strength required to explain away an observed effect, our method shifts the paradigm toward a formal, quantitative sensitivity analysis. In diverse simulations, sensGAN accurately recovers latent structures and outperforms existing methods in identifying confounder-sensitive genes. Applied to human Alzheimer’s disease microglia, our framework prioritizes robust disease pathways while successfully isolating signals driven by unmeasured co-occurring neurodegenerative pathologies. Our method is publicly available, deposited at the GitHub repository [yifanlinz/AD-sensitivity-ICML](https://github.com/yifanlinz/AD-sensitivity-ICML).

1. Introduction

A fundamental challenge in modern genomics is the presence of unmeasured confounding variables that can obscure true biological signals. In high-dimensional single-cell studies, transcriptional changes attributed to a primary disease state may instead reflect unmeasured comorbid processes, environmental exposures, or systemic pathologies that are

incompletely recorded. While conventional differential expression analyses prioritize genes based on statistical significance (Squair et al., 2021), they often fail to quantify the sensitivity of these findings to latent confounders: the more sensitive the findings are to latent confounders, the less robust the discoveries are. Computational frameworks, such as surrogate variable analysis (Leek & Storey, 2007; 2008), provide a single point estimate of latent factors to adjust for residual variation but do not explore the spectrum of hypothetical confounding. This leaves researchers without a metric to determine how “strong” a latent factor would need to be to nullify a discovery.

We argue that high-dimensional genomics requires a shift toward formal sensitivity analysis, mirroring classic statistical approaches like Cornfield’s analysis of smoking and lung cancer (Cornfield et al., 1959). Conceptually, we ask: what level of confounding strength, relative to observed covariates, is required to explain away a gene’s association with a treatment or disease label? This is a novel machine learning problem involving the optimization of “worst-case” latent variables within high-dimensional outcome spaces. By treating confounding as a spectrum rather than a fixed correction, we can distinguish between transcripts that are robust to latent biological processes and those whose significance is likely an artifact of omitted variable bias.

To address this, we introduce **sensGAN** (Sensitivity Generative Adversarial Network), a generative adversarial network for sensitivity analysis that leverages deep learning. SensGAN does not aim to identify a unique causal confounder; rather, it characterizes the space of latent variables that are sufficiently predictive to explain away an observed association. SensGAN learns worst-case latent confounders (also called “unmeasured confounders” in other literature) by constraining their predictive gains on both the treatment and outcome, identifying the weakest confounding strength required to nullify each gene’s association with the disease.

1.1. Scientific relevance

The necessity of sensitivity analysis is most evident in the study of complex disorders like Alzheimer’s disease (AD). A major computational obstacle in uncovering “direct” disease signals is that donors frequently harbor co-occurring

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neurodegenerative diseases (co-pathologies). For instance, in late-onset AD, as few as 31% of cases are described by purely AD-specific signatures, with the remainder displaying complex comorbidity patterns that potentially confound existing single-cell analyses (Robinson et al., 2023).

To illustrate the clinical relevance of understanding the role of co-pathologies, we perform a preliminary analysis of a cohort of 4,168 deceased donors from the National Alzheimer’s Coordinating Center (NACC), all of whom were staged for AD and 5 other co-pathologies (Fig. 1A). Our analysis reinforces a prevailing scientific hypothesis that as primary disease severity increases, donors exhibit an increasing burden of these co-pathologies, which significantly impacts clinical outcomes and donor lifespan (Spina et al., 2021). Our results show that as the severity of AD increases, an increasing percentage of those donors have more co-pathologies (Fig. 1B).

The challenge for computational biologists is that the clinical definitions for these co-pathologies are not static; they undergo major updates roughly every decade as scientific understanding evolves (Mirra et al., 1991; Montine et al., 2012; Nelson et al., 2019). Although many of the cited studies suggest that adjusting for co-occurring neurodegenerative disease is important to uncover the “pure” AD transcriptomic changes, the difficulty lies in the evolving definitions of these co-pathologies (Jack Jr et al., 2024). For instance, Figure 1C summarizes how the standard clinical co-pathology staging criteria evolved over the decades of AD research (Mirra et al., 1991; Braak & Braak, 1991; Montine et al., 2012; Nelson et al., 2019). Given this ever-changing landscape, how do we ensure that our single-cell findings today are robust to possible confounding co-occurring neurodegenerative disease that we might formalize in the future?

With this context, our goal for sensGAN is to provide a “future-looking” framework for differential expression. Rather than relying solely on currently formalized staging, our sensitivity analysis offers two critical advantages:

- **(Goal #1) Make robust discoveries even when important confounders are unmeasured or unknown:** SensGAN enables a sensitivity analysis with respect to latent confounders by explicitly constructing plausible worst-case latent confounders under controlled confounding strength to identify the one that nullifies the largest number of differentially expressed genes. Genes that remain significant under these worst-case scenarios are therefore robust to a broad class of latent confounding effects.
- **(Goal #2) Use the learned confounders to suggest what hidden biology might exist:** SensGAN learns explicit latent confounder representations that may reflect underlying biological processes not yet measured

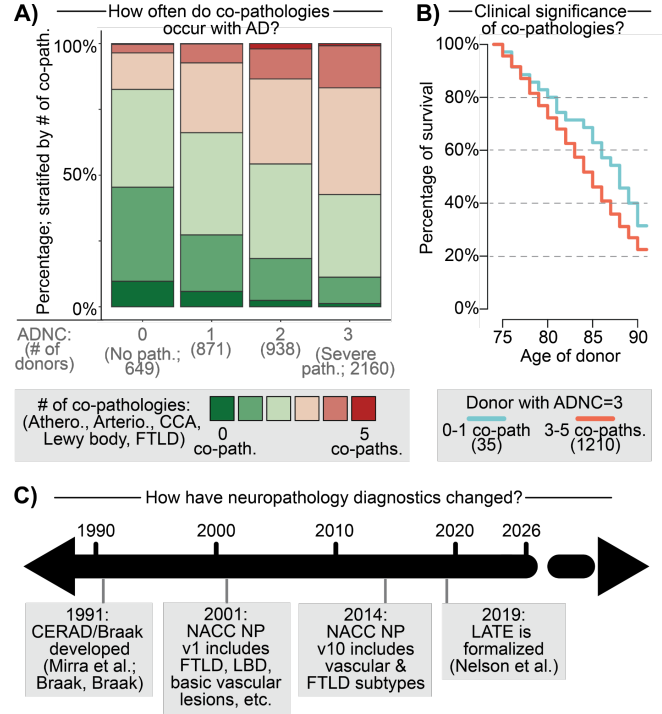


Figure 1. Clinical impact of latent confounders in AD research and sensGAN overview to learn them. **A)** Percentage of donors with number of co-occurring neurodegenerative disease, stratified by AD severity (ADNC score: 0 being no pathologies, 3 being severe). The number of donors in the NACC cohort in each stratum is marked. **B)** Survival curve among donors at age-at-death ≥ 75 with severe AD pathologies, stratified by whether they have 0/1 co-pathologies or 3+ co-pathologies. **C)** Timeline of evolving AD and related co-occurring neurodegenerative diseases were standardized among clinical diagnostics of post-mortem brain tissue.

or formally defined. Although these latent variables are not assumed to correspond to any known pathology, their values can suggest the presence and functional impact of latent biological drivers, providing hypotheses for future experimental or clinical investigation.

1.2. Existing computational methods and their limitations

We model log-normalized, pseudo-bulked gene expression vector $Y \in \mathbb{N}^{n \times p}$ (i.e., the “outcome” across n donors and p genes), treatment status $D \in \{0, 1\}^{n \times 1}$ (i.e., the case-control status), and the d measured covariates $X \in \mathbb{R}^{n \times d}$. Following standard Generalized Linear Model (GLM) frameworks commonly used to model gene expression vector (Sarkar & Stephens, 2021), we assume:

$$Y \sim \text{NB}(\mu, \alpha), \quad \mu = DT_Y + XB_Y + ZT_Y \quad (1)$$

$$D \sim \text{Bernoulli}(\pi), \quad \pi = XB_D + ZT_D$$

Our primary objectives are to estimate the direct effect $T_Y \in \mathbb{R}^{1 \times p}$ of D on Y (Goal #1) and the k -dimensional latent

factor matrix $Z \in \mathbb{R}^{n \times k}$ (Goal #2). Here, $\alpha \in \mathbb{R}_+^p$ denotes gene-specific overdispersions, and $\{B, \Gamma\}$ represent effect matrices for observed and latent variables, respectively, for either the outcome or treatment, depending on the subscript. In practice, Y is often derived from single-cell RNA-seq data, where we aggregate the expression profile across all cells of a particular cell type of interest from each donor, and then log-normalize the expression profile across p highly variable genes of interest.

However, since Z is unobserved, typical DE methods are forced to model the data Y using an “unadjusted” model,

$$\begin{aligned} \hat{Y}_{\text{un}} &\sim \text{NB}(\hat{\mu}_{\text{un}}, \hat{\alpha}_{\text{un}}), \quad \hat{\mu}_{\text{un}} = D\hat{T}_{Y,\text{un}} + X\hat{B}_{Y,\text{un}} \\ \hat{D}_{\text{un}} &\sim \text{Bernoulli}(\hat{\pi}_{\text{un}}), \quad \hat{\pi}_{\text{un}} = X\hat{B}_{D,\text{un}} \end{aligned} \quad (2)$$

We highlight existing computational frameworks below for handling latent confounders Z in this setting, which motivate our proposed method.

Existing frameworks for latent confounding generally fall into two categories. Omitted Variable Bias (OVb) methods quantify the strength required for a latent factor to nullify a treatment-outcome relation (Cinelli & Hazlett, 2020; Veitch & Zaveri, 2020). However, these are largely restricted to linear models and do not scale to the non-linear, high-dimensional outcome spaces (e.g., thousands of genes modeled via Negative Binomial (NB) distributions) typical of single-cell genomics. Conversely, surrogate-variable approaches, such as causarray (Du et al., 2026) and RUVr (Risso et al., 2014), estimate latent factors to adjust for residual variation. These methods are conceptually related to PCA-based approaches, as they seek latent factors that explain as much residual variability as possible after accounting for measured covariates. Yet, these provide only point estimates rather than exploring the spectrum of hypothetical confounding. Furthermore, they often assume that latent factors are independent of the treatment, making them poorly suited to capture neurodegenerative diseases that co-occur with primary disease states like AD. See Appendix B for additional discussion.

2. Overview of sensGAN

Our method, sensGAN, estimates the worst-case latent confounder under constrained confounding strengths using an adversarial framework and provides a sensitivity contour diagnostic plot to suggest more reliable gene targets based on measured variables. In the following, we motivate the critical components of our framework one at a time: (1) quantifying the two “knobs” that define confounding strength, (2) defining the generative adversarial network (GAN) architecture that enables learning the confounder, and (3) training the GAN architecture.

2.1. Predictive gain: defining the confounding strength

We start with our formalization of quantifying the strength of a hypothetical latent confounder, which we denote as $\tilde{Z}$. Intuitively, the strongest latent confounder is the one that improves predictive performance the most for either the genes (Y ; “outcomes”) or the case-control status (D ; treatment). Given $\tilde{Z}$, we can learn $\tilde{\mu}$ and $\tilde{\pi}$ following the generative model in Eq. (1). The predictive gain of $\tilde{Z}$ is defined as the normalized deviance-based partial R^2 (DBPR) defined below, which we denote as R^2 .

For each prediction task (treatment D and outcomes Y), we compare three fitted models: (i) an unadjusted model that does not consider any latent confounders (“un”, see (2)), (ii) an adjusted model using the *strongest* unobserved confounder $\tilde{Z}$ (the k -dimensional confounder that maximizes the likelihood), and (iii) an adjusted model using a particular hypothetical confounder $\tilde{Z}$. Let $\ell_D(\pi)$ and $\ell_Y(\mu, \alpha)$ denote the Bernoulli and negative-binomial (NB) log-likelihoods under (1), and let $\ell_{D,\text{sat}}$ and $\ell_{Y,\text{sat}}$ denote the corresponding *saturated* (perfect-fit) log-likelihoods, i.e., the maximum achievable log-likelihood when each observation has its own free parameter. The deviance of a fitted model is $\text{Dev} = 2(\ell_{\text{sat}} - \ell)$, with smaller deviance meaning a better fit. Dev_D is defined as the sum of Bernoulli deviances across all donors, and Dev_{Y_i} is defined as the sum of NB deviances across all genes. Details and derivations provided in Appendix C. The deviance-based partial R^2 for including Z into the model is the fraction of unadjusted deviance removed:

$$R_D^2 = \frac{\text{Dev}_{\hat{D}_{\text{un}}} - \text{Dev}_D}{\text{Dev}_{\hat{D}_{\text{un}}}}, \quad R_{Y_i}^2 = \frac{\text{Dev}_{\hat{Y}_{\text{un}},i} - \text{Dev}_{Y_i}}{\text{Dev}_{\hat{Y}_{\text{un}},i}}.$$

Finally, we normalize by the maximal gain achieved by $\hat{Z}$ to obtain bounded predictive-gain “knobs” $\kappa, \eta \in [0, 1]$, defined as

$$\kappa = \frac{R_D^2}{R_{\hat{D}}^2}, \quad \eta = \frac{\frac{1}{n} \sum_{i=1}^n R_{Y_i}^2}{\frac{1}{n} \sum_{i=1}^n R_{\hat{Y}_i}^2}, \quad (3)$$

which quantify how predictive $\tilde{Z}$ is relative to the strongest confounder in our model class. These emulate similar properties and interpretations of partial R^2 for linear models as discussed in sensmarker (Cinelli & Hazlett, 2020). The normalized predictive gain denotes the proportional predictive gain introduced by an arbitrary $\tilde{Z}$, normalized by the maximal predictive gain introduced by the most powerful $\hat{Z}$. A normalized predictive gain of 1 means that $\tilde{Z}$ achieves 100% of the maximal predictive gain. Importantly, it is also differentiable, allowing gradient calculation in a deep learning model, as we describe later in this section.

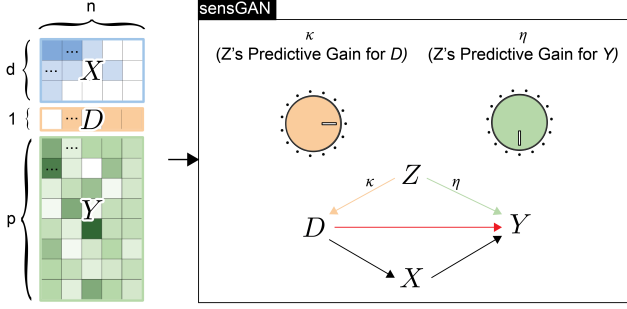


Figure 2. sensGAN core idea. As illustrated, unobserved confounders may simultaneously influence both the treatment D and the outcome Y , leading to spurious associations. sensGAN parameterizes this confounding strength using two normalized predictive-gain knobs and systematically explores the resulting confounding scenarios.

2.2. Architecture and loss

Now that we have a quantification of the impact of a latent confounder, we next need to describe sensGAN’s deep-learning architecture, which allows us to learn the confounder Z that meets our specifications. Broadly speaking, sensGAN’s architecture has two fundamental components: the predictor and the generator. Here, we first describe each component of the architecture as shown in Fig. 3.

Predictors. The predictors $f_D(\cdot)$ and $f_Y(\cdot)$ are 1-layer (i.e., shallow) fully connected networks that estimate model parameters by maximizing the likelihoods. To D_i , the treatment predictor $f_D(\{X_i, Z_i\})$ uses a binary cross-entropy (BCE) loss. To predict Y_i , the outcome predictor $f_Y(\{D_i, X_i, Z_i\})$ uses a NB log-likelihood loss. Both predictors are designed to emulate logistic regression and NB GLM regression, respectively. There are no activation functions involved, so the weight corresponding to variable D in $f_Y(\cdot)$ can be interpreted as the “coefficient” T_Y in our generative model; see Eq. (1). The overdispersion parameters for modeling Y are estimated from the unadjusted NB model fitting.

Generator. The generator $g(\cdot)$ deterministically maps donor-level information to k latent confounders $\tilde{Z}_i \in [0, 1]^k$ for donor i , based on covariates X_i , treatment status D_i , and outcome (gene expression) vector Y_i . By default, the generator is a two-layer fully connected neural network with ReLU activation that produces $\tilde{Z}_i$ via a sigmoid output layer.

2.3. Training

Equipped with the predictor and generator architectures, we are now ready to explain how we train sensGAN. Drawing inspiration from the OVB literature (Cinelli & Hazlett, 2020; Veitch & Zaveri, 2020), we strive to learn latent confounders $\tilde{Z}$ that nullifies as many genes (i.e., eliminates the

significance of the largest number of genes associated with D), such that $\tilde{Z}$ has a particular pre-specified predictive gain $\{\kappa, \eta\}$. The overall procedure has two main steps:

- **Step 1:** Learn the unadjusted model and the most powerful confounder, both of which are needed to define the normalized DBPRs; see Eq. (3),
- **Step 2:** Train the predictors $f_D(\cdot)$ and $f_Y(\cdot)$ and generator $g(\cdot)$ in an adversarial manner.

We highlight each step below. The full pseudocode is described in Appendix G.

Step 1: Unadjusted predictors and most powerful confounder. To operationalize our definition of the predictive gains in Eq. (3), we need to first compute the likelihood of the unadjusted model and the adjusted model using the most predictive confounder $\hat{Z}$.

To learn the unadjusted models, we fit a logistic regression to predict D from X , and fit p NB GLMs to predict each gene in Y from X and D . From these fitted models, we can compute $\text{Dev}_{\hat{D}_{\text{un}}}$ and $\text{Dev}_{\hat{Y}_{\text{un}}}$, respectively. The gene-specific overdispersion parameters α estimated at this stage are held fixed and used throughout all subsequent sensGAN calculations.

We then train the predictors and the generator to minimize the joint negative log-likelihood, yielding the most powerful confounder $\hat{Z}$ and the corresponding models $f_D(\cdot)$, $f_Y(\cdot)$, and $g(\cdot)$. Specifically, we define the BCE loss based on predicted logits as $\mathcal{L}_{\text{BCE}}$ and the NB loss based on the predicted mean as $\mathcal{L}_{\text{NB}}$. Predictors are trained respectively by minimizing $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{NB}}$. Generators are trained by minimizing

$$\mathcal{L}_L = \lambda_{\text{BCE}} \cdot \mathcal{L}_{\text{BCE}} + \lambda_{\text{NL}} \cdot \mathcal{L}_{\text{NB}},$$

where $\lambda_{\text{BCE}}, \lambda_{\text{NL}} \geq 0$ are hyperparameters controlling the weight of their respective terms. From this fitted model, we can compute $\text{Dev}_{\hat{D}}$ and $\text{Dev}_{\hat{Y}}$, respectively. More details are discussed in Appendix E.

Step 2: Adversarial training We now describe how we train predictors $f_D(\cdot)$ and $f_Y(\cdot)$ and generator $g(\cdot)$. After initializing the GAN architecture (see Appendix F), we then update the predictors and generator alternatively and adversarially – the predictors maximize log-likelihoods while the generator controls for predictive gains. Specifically, in one iteration, the generator is frozen and the predictors $f_D(\cdot)$ and $f_Y(\cdot)$ are updated; in the other, the predictors are frozen and the generator $g(\cdot)$ is updated.

When updating the predictors, we minimize $\mathcal{L}_{\text{BCE}}$ to update the weights in $f_D(\cdot)$ and minimize $\mathcal{L}_{\text{NB}}$ to update the weights in $f_Y(\cdot)$. This is relatively straightforward, as both

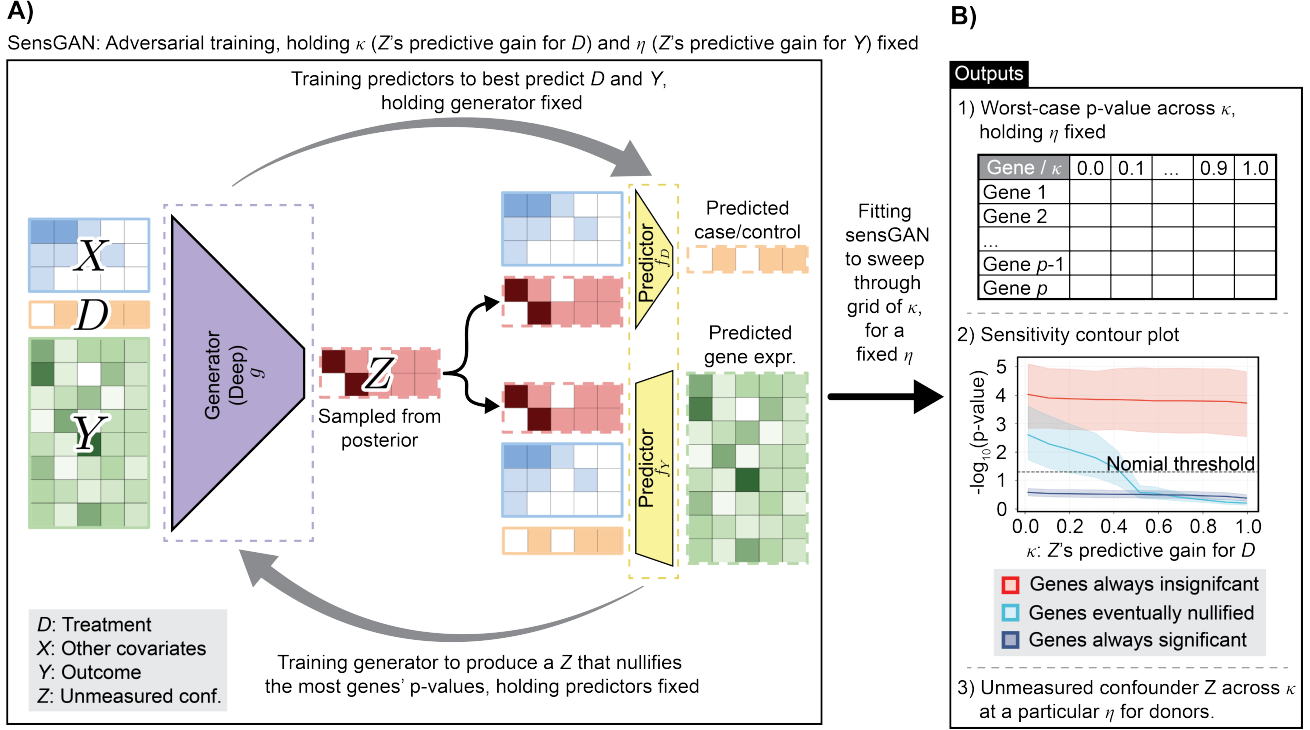


Figure 3. **sensGAN method overview.** **A)** The GAN system that learns the confounder Z with target predictive gains (κ, η) , achieving predictive gains (κ', η') . **B)** Method outputs while sweeping through a grid of κ and holding η fixed.

predictors are designed to emulate logistic and NB GLMs, respectively, where the optimization is convex.

Generator optimization is more involved and represents a key component of the sensGAN framework. The loss function when optimizing $g(\cdot)$ involves three main terms: the plausibility objective $\mathcal{L}_P$, the significance objective $\mathcal{L}_S$, and a regularizer $\mathcal{L}_{\text{reg}}$. We explain each component below.

- To motivate the plausibility objective, we note that the strongest confounder $\tilde{Z}$ explains away all unexplained residuals in D and Y , by construction. This is unlikely to reflect any practical latent confounder. We thus define the *plausibility objective* ($\mathcal{L}_P$): the predictive gains of plausible confounders $\tilde{Z}$ with respect to D and Y are constrained by prespecified targets, $(\kappa, \eta) \in [0, 1]$,

$$\mathcal{L}_P = \lambda_Y \cdot \left[((\eta' - \eta) s_Y)^2 \right] + \lambda_D \cdot \left[((\kappa' - \kappa) s_D)^2 \right],$$

where κ' and η' denote the achieved predictive gains with respect to D and Y , respectively; s_Y and s_D are scaling factors, and $\lambda_Y, \lambda_D \geq 0$ are hyperparameters.

- To motivate the significance objective, we formalize “worst-case confounder” given κ and η as the confounder $\tilde{Z}$ that nullifies the largest number of significant genes. Thus, we define the *significance objective* ($\mathcal{L}_S$) that minimizes the average sigmoid-transformed excess of the test statistics over the significance threshold $\delta \in (0, 1)$ (i.e., typically

$\delta = 0.05$),

$$\mathcal{L}_S = \lambda_{\text{pval}} \cdot \frac{1}{p} \sum_{j=1}^p \sigma_{\lambda} \left(\left| \frac{[\tilde{T}_Y]_j}{\text{SE}([\hat{T}_{Y,\text{un}}]_j)} \right| - z_{1-\delta} \right),$$

where $[\hat{T}_{Y,\text{un}}]_j$ and $[\tilde{T}_Y]_j$ denote the estimated coefficients of D for gene j in the unadjusted and adjusted NB models, see Eq. (2) and (1) respectively. $\text{SE}(\hat{T}_{Y,\text{un},j})$ is the corresponding standard error from the unadjusted model. To compute the test statistic, we approximate $\text{SE}(\hat{T}_{\text{un},j}) \approx \text{SE}(\tilde{T}_j)$, assuming comparable uncertainty before and after adjustment. $z_{1-\delta}$ is the $(1 - \delta)$ quantile of the standard normal distribution. $\sigma_{\lambda}(\cdot)$ denotes a sigmoid function, which smoothly maps test statistics onto the unit interval. $\lambda_{\text{pval}} \geq 0$ is a hyperparameter controlling the overall strength of the significance penalty.

- The last term is the *regularizer* ($\mathcal{L}_{\text{reg}}$). This includes a correlation regularizer, a cosine similarity loss, and a variance-matching penalty, all of which we have empirically found help stabilize the learning of the latent confounder Z and prevent the model from collapsing to a constant value. See more details in Appendix E.

Combining all the loss terms together, the objective function of the generator $g(\cdot)$ given pre-specified (and fixed) predictive gains κ^* and η^* is to minimize,

$$\mathcal{L}_S + \mathcal{L}_P + \mathcal{L}_{\text{reg}}.$$

To span all plausible confounders, we train the sensGAN model sequentially by sweeping over the predictive gains (κ, η) on the unit interval $[0, 1]$. At the end of the sequential training, the generator finds a series of worst-case confounders with varied predictive gains.

The latent confounders Z can be used in a standard NB GLM analysis where we regress $\{X, D, Z\}$ onto Y , where we record the significance of each gene’s coefficient in T_Y , see Eq. (1). Gene-level p-values are computed using Wald statistics, where the test statistic for each gene is formed by dividing the adjusted coefficient by its standard error estimated from the unadjusted model $(\widehat{T}_Y)_j / \text{SE}(\widehat{T}_{Y,\text{un}})_j$. Mathematically, the p-value of each gene changes in the model that adjusts for our learned latent confounder, compared to the unadjusted model, see Eq. (2)). We are interested in the specific value of κ^* that “nullifies” the gene given a fixed η , if it is nullified in the adjusted model.

Additional details, such as the choice of hyperparameters, are deferred to Appendix F. Given the potential instability of GAN training as mentioned by anonymous reviewers, we also discuss an alternative training strategy using bilevel optimization, where we elaborate on it in Appendix F.3. Furthermore, we elaborate further on the modeling strategy, explaining why we chose to model scRNA-seq data as pseudobulked samples rather individual cells, as done in work like causarray (Du et al., 2026) and CoCoA-diff (Park & Kellis, 2021), in Appendix D.

2.4. Downstream analysis: Calibration of predictive gains with measured covariates

From the sensitivity analysis, sensGAN produces a series of plausible worst-case confounders indexed by normalized predictive gains (κ, η) with respect to the exposure D and outcome Y . To interpret these latent confounders relative to observed covariates, we perform a calibration step that computes predictive gains for measured covariates, placing latent and measured effects on a common scale. If a worst-case confounder with strength x times that of the strongest measured covariate is required to nullify a gene’s significance, large values of x indicate that such confounding is potentially implausible, supporting the robustness of the discovery. See more details in Appendix F.

2.5. Downstream analysis: Diagnostic sensitivity contour plots

To connect sensGAN’s core idea to gene-level robustness, we introduce a *sensitivity diagnostic* inspired by (Cinelli & Hazlett, 2020). Fixing the predictive-gain knob for the outcome (η) to reflect a conservative but bounded amount of confounding in the outcome, we vary the treatment-side knob (κ) and record the corresponding worst-case p-values. This assesses how predictive a latent confounder needs to be

of D in order to nullify a gene’s significance. Genes that lose significance at small κ are sensitive to weak confounding, whereas genes that remain significant until large κ exhibit greater robustness.

3. Experimental Setup

3.1. Analysis of the simulation dataset for method comparison

We evaluate sensGAN on simulated data generated from a model with a single unobserved binary confounder ($n=100$, $p=100$, $d=4$, $k=1$). Observed covariates X and the binary latent confounder Z jointly influence a case-control variable D through a logistic regression model, while the outcome matrix Y is generated from a NB generalized linear model depending on X , D , and Z . By varying the effects of D and Z on gene expression, we simulate three groups of genes: (i) genes truly associated with D but not Z , (ii) genes associated with Z but not D , and (iii) genes associated with neither. More details and parameter settings are provided in Appendix H.

To evaluate different methods, we first compare the correlations between the learned confounder and the true confounder (Fig. 4). The most powerful confounder identified by sensGAN shows a correlation comparable to that of causarray (Du et al., 2026) and RUVr (Risso et al., 2014), two surrogate-variable methods that we benchmark against. As the predictive-gain knob, κ , is turned from 1 to 0, the correlation between the learned and true confounders decreases smoothly from approximately 0.75 to 0.4. sensGAN not only identifies the strongest latent confounding structure, on par with competing approaches, but also generates a continuum of plausible worst-case confounders constrained by predictive gains.

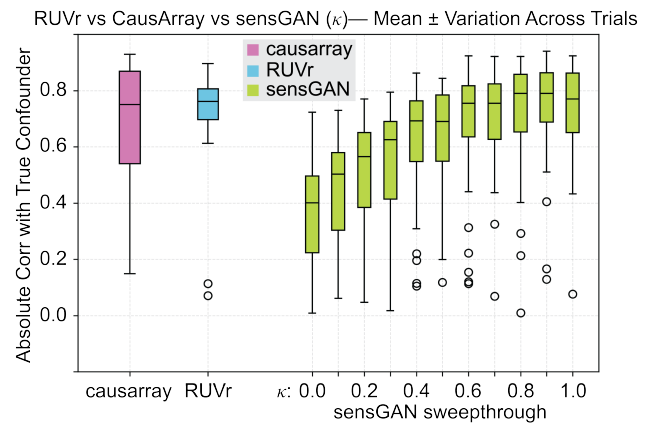


Figure 4. Correlation comparisons among methods. Correlations between the true confounder and the confounders estimated by causarray, RUVr, and sensGAN.

Next, we compare contingency tables of gene significance

Table 1. Comparison of mean gene counts across simulated significance categories. Rows: true gene categories, columns: inferred categories assigned by each method; values report mean counts averaged over 10 trials.

Method	Category	# Sig.	# Null.	# Insig.
Unadjusted	Significant	12.64	–	2.36
	Nullified	10.64	–	3.36
	Insignificant	4.06	–	64.94
causarray	Significant	13.18	–	1.82
	Nullified	6.88	–	7.12
	Insignificant	31.70	–	37.30
RUVr	Significant	11.04	–	3.96
	Nullified	6.92	–	7.08
	Insignificant	4.42	–	64.58
sensGAN	Significant	12.04	0.76	2.20
	Nullified	2.40	8.14	3.46
	Insignificant	2.20	3.90	62.90

across methods (Table 1). All methods correctly identify always-significant genes, while causarray behaves conservatively for always-insignificant genes, likely due to its design for single-cell rather than donor-level pseudobulk data. Differences are most pronounced for nullified genes: the unadjusted GLM naively labels all such genes as significant (Fig. 4B), and causarray and RUVr split them between significant and insignificant categories. In contrast, sensGAN learns a family of plausible confounders with varying predictive gains and identifies a substantial fraction of nullified genes under worst-case confounding (8 out of 15; Table 1), while maintaining high accuracy for always-significant and always-insignificant genes (Fig. 5).

We further present the sensitivity diagnostic contour plot based on sensGAN outputs (Fig. 4). We focus on nominal p-values rather than multiple-testing-adjusted p-values, as these allow us to better investigate how p-values change as the level of confounding increases. The y-axis is the logarithm of the worst-case p-values, the black dashed line is the significance threshold of 0.05, and the x-axis is the predictive knob (κ) for the treatment. Every gene has a contour curve that depicts its sensitivity to the estimated latent confounder, exemplified by the thin curves. The thick curves are the mean contour curves, and the bands are the 95% confidence intervals, colored by the simulated gene categories. If a gene crosses the significance threshold with a smaller κ , the gene is considered to be nullified by a weaker confounder. Compared to genes nullified by stronger confounders, this gene is less directly associated with the treatment. Therefore, the differential expression conclusion is less robust.

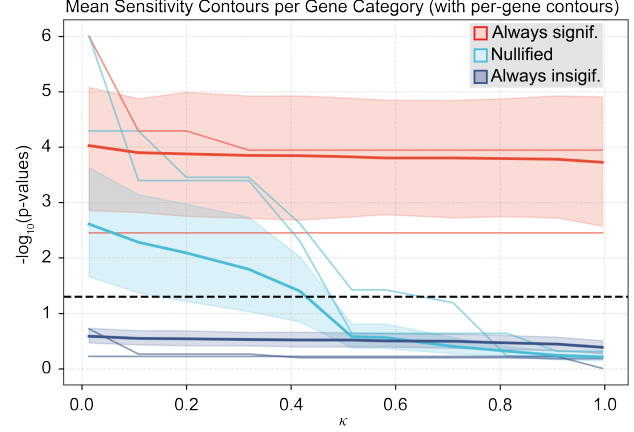


Figure 5. Simulation demonstrates sensGAN’s accuracy. Sensitivity diagnostic contour plot With fixed $\eta = 0.5$, the thick contour curve represents the mean across genes, the thin curve shows an individual gene for illustration, and the shaded band denotes the 95% confidence interval.

3.2. Validation of contour curve accuracy

Having shown that sensGAN recovers latent confounders correlated with those identified by existing methods, we next evaluate its ability to accurately characterize sensitivity contour curves, which prior approaches do not provide. Our primary focus is to investigate whether sensGAN can faithfully track how gene-level p-values evolve with increasing confounding strength. To this end, we simulate data as before ($n = 100$, $p = 104$, $d = 4$, $k = 1$), but consider a continuous confounder Z to enable finer control over predictive gains, and restrict attention to genes that are associated with Z but not with D (Category (ii)). We first construct reference sensitivity contour curves using a crawler-based search that manually explores the predictive-gain space to identify the “true” worst-case confounder for a particular setting of (κ, η) (see Appendix H). We then apply sensGAN to the same simulated data and compare the resulting sensitivity contours, assessing whether sensGAN accurately reproduces the true nullification behavior across predictive-gain levels.

We first evaluate sensGAN at the gene level by comparing per-gene sensitivity contour curves against the reference contours (Fig. 5), with a primary focus on the difference in κ^* at the nullification point. Quantitatively, the difference is modest, with a mean difference of 0.05 and a 95% confidence interval of (0.00, 0.22), indicating close alignment in the predictive-gain level at which genes lose significance.

We next compare the mean sensitivity contour curves across genes (Fig. 6B). Overall, the mean curves produced by sensGAN closely track the reference contours, with similar shapes and comparable crossing points near $\kappa \approx 0.95$. The 95% confidence intervals largely overlap across the

predictive-gain range, suggesting that sensGAN captures the dominant trend in how significance decays with increasing confounding strength. See Appendix I for more results.

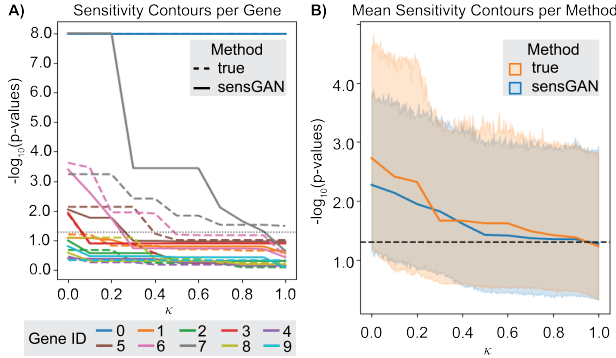


Figure 6. Sensitivity contour plot comparison between the reference and the sensGAN output. A) Per-gene sensitivity contour plots showing $-\log_{10}(p)$ as a function of the predictive-gain parameter κ for individual genes. Dashed curves denote reference contours obtained via the crawler, while solid curves denote contours recovered by sensGAN. B) Mean sensitivity contour curves across genes, with shaded regions indicating 95% confidence intervals.

3.3. Analysis of microglia studying Alzheimer’s disease, adjusting for unmeasured co-occurring neurodegenerative disease

To demonstrate how sensGAN can clarify the biological signals, we showcase our analysis of systemic lupus erythematosus (SLE) and Alzheimer’s disease (AD). We discuss the AD case-study here in detail, while deferring the SLE case-study to Appendix J. For our AD case-study, we leverage the Seattle Alzheimer’s Disease Brain Cell Atlas (SEA-AD) consortium (Gabbito et al., 2024) and specifically focus on the microglia in the prefrontal cortex (PFC) due to its critical role in clearing AD pathology and its implication from established GWAS studies (Karch & Goate, 2015; Scheltens et al., 2016).

A major challenge in studying cellular changes during AD progression is that post-mortem donors often exhibit multiple neurodegenerative diseases beyond AD itself (Nichols et al., 2023; McAleese et al., 2021). These co-occurring diseases (“co-pathologies”) can confound single-cell analyses. For example, only 31% of late-onset AD cases show AD-specific signatures (Robinson et al., 2023).

We construct a pseudobulk gene expression matrix per donor (i.e., Y) and perform our sensGAN analysis to determine which genes are impacted based on whether the donor has AD pathology after post-mortem dissection of their brain tissue (i.e., D), adjusting for sex, age, APOE4 status, ethnicity, and post-mortem interval (PMI) (i.e., X). Since we are only interested in whether or not significant genes remain

significant during our sensitivity analysis, we focus on a key set of genes that have a $p\text{-value} < 0.2$ when doing a differential expression test via DESeq2 (Love et al., 2014), resulting in $p = 448$ genes in our analysis. Our goal is to determine which nominally significant genes are nullified by the latent confounders identified by sensGAN (Goal #1) and what these latent confounders represent biologically (Goal #2).

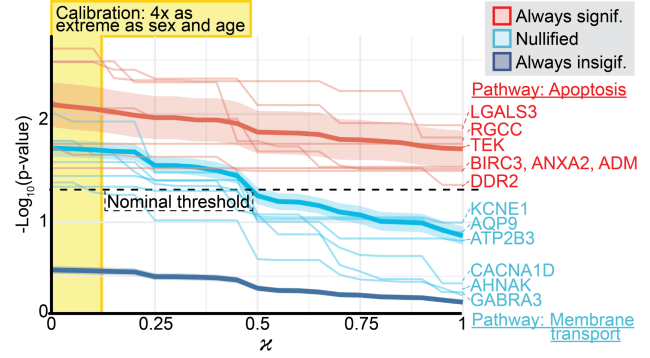


Figure 7. sensGAN prioritizes DEGs specific to microglia in AD. Diagnostic contour, showing genes that are always significant (red), nullified (light blue), and always insignificant (dark blue) across different values of κ . The calibration with respect to age and sex is shown (yellow).

Our sensGAN analysis, sweeping across κ from 0 to 1, disentangles potential microglial pathways that may be confounded by co-pathologies from those that are specific to AD. Our diagnostic contour plots reveal that genes that are always nominally significant are enriched for an apoptosis pathway (red; Fig. 7) (Dou et al., 2024). For instance, the *LGALS3* gene is strongly upregulated in plaque-associated microglia in AD models and human tissue (Tan et al., 2021), and has been implicated to be AD-specific through knock-outs in AD mouse models, which demonstrated reduced plaque burden and improved cognitive behaviors (Siew et al., 2024). In stark contrast, genes that are eventually nullified by sensGAN’s learned latent confounders Z are enriched for transmembrane transport (light blue; Figure 7). For instance, *AHNK* regulates voltage-gated calcium channels (Matza et al., 2008). Although it is nominally significant without confounder adjustment, this gene is quickly nullified by sensGAN. This is plausible, since its association with AD is not solely through microglia but also through its interaction with neurons (Wang et al., 2025). Furthermore, it has also been suggested to be involved in Lewy bodies (Santpere et al., 2018) and Frontal Temporal Dementia (Lorenzini et al., 2023), two other co-occurring pathologies typically associated with AD. Additionally, *AQP9*, a gene that regulates membrane channels to conduct water, is also nullified, which is canonically associated with an AD-relevant gene for astrocytes, not microglia (Liu et al., 2018). Together, these results demonstrate that sensGAN can meaningfully “purify”

DEG results, retaining only microglia-specific DEGs robust to non-AD latent confounders.

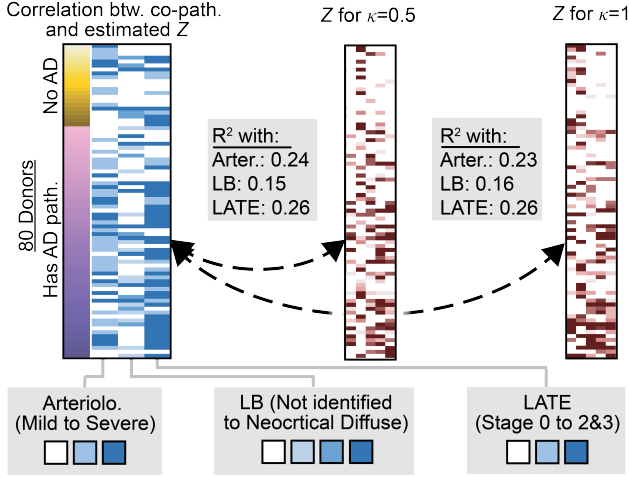


Figure 8. sensGAN’s latent confounders reflect co-pathologies. Donors (rows) and their measured co-pathology, as well as the estimated latent confounders for $\kappa \in \{0.5, 1\}$.

Next, we illustrate that sensGAN’s latent confounders are suggestive of currently measured co-occurring neurodegenerative disease. While we do not use any co-pathologies in our sensGAN analysis, the SEA-AD consortium also stages all the post-mortem tissues for arteriolosclerosis (i.e., a vascular disease), Lewy bodies, and Limbic-predominant age-related TDP-43 encephalopathy (LATE). We compute how well sensGAN’s learned confounders predict these co-pathologies (Fig. 8). Surprisingly, for $\kappa \in [0.5, 1]$, the predictive power measured by R^2 between the predicted (linear regression) and observed co-pathology stages is significantly non-zero and not changing. This suggests that sensGAN latent confounders are capturing co-pathology impacts from snRNA-seq data, even with a modest value of $\kappa = 0.5$. We hypothesize that the latent confounders estimated by sensGAN additionally capture effects beyond co-pathology, for instance, the effects of spatial microenvironments, genetics, or environmental factors (e.g., diet) on the donor that affect AD.

We further examine which microglial states harbor the DEGs identified by sensGAN. Although sensGAN operates on pseudobulk data, mapping DEGs to microglial states provides biological context (Prater et al., 2023). We first visualize the microglia in the SEA-AD cohort by their annotated state (Fig. 9A). Significant genes that are either never nullified by sensGAN (*LGALS3*, *ANXA2*) or eventually nullified (*AQP9*, *AHNAK*) show elevated expression primarily in DAM and transitory microglia (Fig. 9B). These results highlight how sensGAN can prioritize DEGs based on their robustness to latent confounding while preserving biologically meaningful microglial localization.

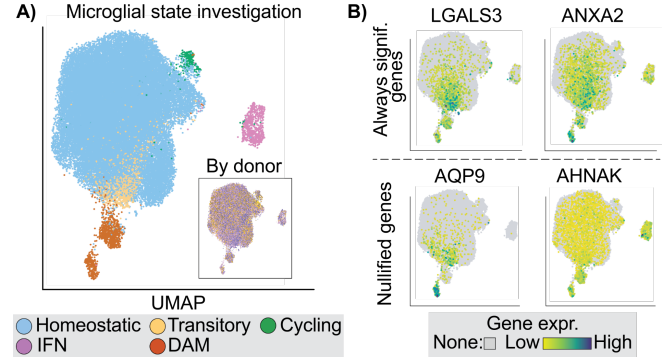


Figure 9. sensGAN prioritizes DEGs specific to microglia in AD. A) UMAP of single-nuclei microglia in SEA-AD, colored by microglial state and donor. B) Expression of certain genes across the microglia.

3.4. Discussion

We present sensGAN, an adversarial framework that shifts the focus of high-dimensional genomic analysis from binary significance to quantitative robustness. Unlike traditional surrogate-variable methods that provide single-point estimates of latent factors, sensGAN explicitly explores the confounding spectrum by learning “worst-case” latent variables under controlled predictive-gain constraints. Our simulations demonstrate that sensGAN maintains high fidelity in separating robust biological signals from those easily nullified by hidden confounders, effectively capturing a continuum of plausible confounding scenarios that point-estimation methods often miss. By treating confounding as a formal machine learning optimization problem, sensGAN provides a principled metric for determining how strong a latent process must be to explain away a discovery.

Applied to complex diseases, sensGAN successfully “purifies” differential expression results, isolating pathways specifically associated with AD from signals likely driven by co-occurring neurodegenerative disease. For instance, our framework prioritizes apoptosis-related processes as robust AD-specific drivers, while identifying transmembrane transport genes as highly sensitive to unmeasured co-occurring neurodegenerative disease, such as Lewy body pathology. While currently limited by its reliance on pseudobulk aggregation and calibrated (rather than absolute) effect sizes, sensGAN establishes a unified, future-oriented approach to interpreting single-cell genomics in the presence of pervasive, evolving confounding.

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Impact statement

This work introduces an adversarial framework to quantify the robustness of high-dimensional genomic findings against latent confounding. By transitioning from binary significance to a quantitative exploration of the confounding spectrum, this method facilitates more reliable interpretation of biological mechanisms and the prioritization of robust therapeutic targets. This advancement enhances the reproducibility and trustworthiness of computational biology, with potential long-term societal benefits for precision medicine across diverse complex diseases.

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A. Code and data availability

The sensGAN method is publicly available at https://github.com/yifanlinz/AD_sensitivity_ICML, which includes both the implementation of the method and the way it is used on different datasets in this work.

The systemic lupus erythematosus data (Perez et al., 2022), which we analyze in Section J, is publicly available and can be downloaded at <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE174188>. The Alzheimer’s dataset (Gabbito et al., 2024) is publicly available and can be downloaded from <https://registry.opendata.aws/allen-sea-ad-atlas/>, and we focus on analyzing the microglial data from the prefrontal cortex.

B. Motivation for confounder modeling

The challenge of latent confounding is particularly acute in Alzheimer’s Disease (AD) research, where donors frequently exhibit complex comorbidity patterns or co-pathologies. Current computational methods struggle to isolate “direct” disease signals from these co-occurring processes.

Comparison with Omitted Variable Bias (OVb) framework. Methods such as sensmaker (Cinelli & Hazlett, 2020) utilize partial R^2 to reparameterize bias in linear models. While foundational, these approaches are not easily generalized to the non-linear Negative Binomial (NB) distributions required for pseudobulked single-cell RNA-seq data. sensGAN addresses this gap by enabling a formal sensitivity analysis across thousands of concurrent outcomes (genes) within a deep-learning adversarial framework.

Comparison with Surrogate Variable Analysis (SVA) : Standard genomic adjustment tools, including RUVr (Risso et al., 2014), CATE (Wang et al., 2017), GCATE (Du et al., 2025), and causarray (Du et al., 2026), focus on capturing residual variation through low-dimensional surrogates. While these provide explicit latent representations, they typically treat confounding as a fixed correction rather than a spectrum. Because these estimates are often independent of the treatment variable, they may fail to reflect biologically relevant latent factors—such as vascular disease or Lewy body pathology—that are statistically associated with both the AD case-control status and the observed gene expression matrix.

C. Details on deviance, DBPR, and normalized predictive gains

In Section C.1, we describe the four different models used in the calculation of the DBPR. In Sections C.2 and C.3, we formally define deviance and DBPR. In Section C.4, we formally define the “knobs” κ and η used in sensGAN to quantify the predictive gain of a latent confounder.

C.1. Model family and fitted quantities.

For donors $i \in \{1, \dots, n\}$, let $D_i \in \{0, 1\}$ denote treatment (i.e., case-control status), $X_i \in \mathbb{R}^d$ observed covariates, and Y_{ij} the (pseudobulk) count for gene $j \in \{1, \dots, p\}$. For any candidate latent confounder matrix $Z \in \mathbb{R}^{n \times k}$, our generative family induces (i) a Bernoulli model for D with fitted linear predictor π (notation as in the main text), and (ii) a negative-binomial (NB) model for Y with fitted mean μ and gene-specific dispersion α . We write the corresponding log-likelihoods as $\ell_D(\pi)$ and $\ell_Y(\mu, \alpha)$.

We instantiate the models in four standard ways for each task:

1. **Saturated model (sat):** The saturated models are the *perfect-fit* references that achieve the maximum possible log-likelihood. Denote these maxima by $\ell_{D,\text{sat}}$ and $\ell_{Y,\text{sat}}$.
2. **Unadjusted model (un):** Fit models *without* Z to obtain $(\hat{\pi}_{\text{un}}, \hat{\mu}_{\text{un}}, \hat{\alpha}_{\text{un}})$ and log-likelihoods $\ell_D(\hat{\pi}_{\text{un}})$, $\ell_Y(\hat{\mu}_{\text{un}}, \hat{\alpha}_{\text{un}})$. In the following models, overdispersion is fixed at $\hat{\alpha}_{\text{un}}$.
3. **Most predictive confounder (Denoted by $\hat{\cdot}$):** Define $\hat{Z}$ as a confounder of dimension- k that yields the best achievable fit when included in the models with unadjusted overdispersion α , i.e., it maximizes the likelihood (equivalently minimizes deviance). After fitting, this produces $(\hat{\pi}, \hat{\mu})$ and the log-likelihoods $\ell_D(\hat{\pi})$, $\ell_Y(\hat{\mu}, \hat{\alpha}_{\text{un}})$.
4. **Arbitrary confounder (Denoted by $\tilde{\cdot}$):** For any specific confounder $\tilde{Z}$ (e.g., produced by sensGAN under constraints), fit models with the unadjusted overdispersion α including $\tilde{Z}$ to obtain $(\tilde{\pi}, \tilde{\mu})$ and the log-likelihoods $\ell_D(\tilde{\pi})$, $\ell_Y(\tilde{\mu}, \hat{\alpha}_{\text{un}})$.

C.2. Deviance

For modeling D , the Bernoulli deviance based on logits $\text{Dev}_D \in \mathbb{R}$ is defined:

$$\text{Dev}_{\hat{D}_{\text{un}}} = \sum_{i=1}^n 2(\ell_{D_i, \text{sat}} - \ell_{D_i}(\hat{\pi}_{\text{un}})), \quad \text{Dev}_{\hat{D}} = \sum_{i=1}^n 2(\ell_{D_i, \text{sat}} - \ell_{D_i}(\hat{\pi})), \quad \text{Dev}_{\tilde{D}} = \sum_{i=1}^n 2(\ell_{D_i, \text{sat}} - \ell_{D_i}(\tilde{\pi})).$$

For modeling Y , the NB deviance $\text{Dev}_Y = \{\text{Dev}_{Y_1}, \text{Dev}_{Y_2}, \dots, \text{Dev}_{Y_n}\} \in \mathbb{R}^n$ is the sum across genes:

$$\begin{aligned} \text{Dev}_{\hat{Y}_{\text{un}, i}} &= \sum_{j=1}^p 2(\ell_{Y_{ij}, \text{sat}} - \ell_{Y_{ij}}(\hat{\mu}_{ij, \text{un}}, \hat{\alpha}_{j, \text{un}})), \\ \text{Dev}_{\hat{Y}_i} &= \sum_{j=1}^p 2(\ell_{Y_{ij}, \text{sat}} - \ell_{Y_{ij}}(\hat{\mu}_{ij}, \hat{\alpha}_{j, \text{un}})), \\ \text{Dev}_{\tilde{Y}_i} &= \sum_{j=1}^p 2(\ell_{Y_{ij}, \text{sat}} - \ell_{Y_{ij}}(\tilde{\mu}_{ij}, \hat{\alpha}_{j, \text{un}})). \end{aligned}$$

Thus, deviance behaves like a likelihood-based error metric: lower deviance indicates a better fit.

C.3. DBPR: deviance-based partial R^2

We quantify the *incremental predictive gain* ($R_D^2 \in \mathbb{R}$, $R_Y^2 = \{R_{Y_1}^2, R_{Y_2}^2, \dots, R_{Y_n}^2\} \in \mathbb{R}^n$) of adding a confounder Z by the fraction of unadjusted deviance it removes.

Using the fitted models above, this yields the quantities shown in the main text:

$$\hat{R}_D^2 = \frac{\text{Dev}_{\hat{D}_{\text{un}}} - \text{Dev}_{\hat{D}}}{\text{Dev}_{\hat{D}_{\text{un}}}}, \quad \hat{R}_{Y_i}^2 = \frac{\text{Dev}_{\hat{Y}_{\text{un}, i}} - \text{Dev}_{\hat{Y}_i}}{\text{Dev}_{\hat{Y}_{\text{un}, i}}}, \quad \tilde{R}_D^2 = \frac{\text{Dev}_{\hat{D}_{\text{un}}} - \text{Dev}_{\tilde{D}}}{\text{Dev}_{\hat{D}_{\text{un}}}}, \quad \tilde{R}_{Y_i}^2 = \frac{\text{Dev}_{\hat{Y}_{\text{un}, i}} - \text{Dev}_{\tilde{Y}_i}}{\text{Dev}_{\hat{Y}_{\text{un}, i}}}.$$

C.4. Normalized DBPR (predictive-gain knobs)

Because $\hat{Z}$ is defined to be the *most predictive* confounder in our model class, $\hat{R}_D^2$ and $\hat{R}_Y^2$ represent maximal attainable deviance reduction for predicting D and Y , respectively. We therefore define normalized predictive gains for any $\tilde{Z}$ by

$$\kappa = \frac{\tilde{R}_D^2}{\hat{R}_D^2}, \quad \eta = \frac{\frac{1}{n} \sum_{i=1}^n \tilde{R}_{Y_i}^2}{\frac{1}{n} \sum_{i=1}^n \hat{R}_{Y_i}^2},$$

so that $\kappa, \eta \in [0, 1]$ quantify the *fraction of maximal predictive gain* achieved by $\tilde{Z}$ for D and Y .

D. SenGAN architectures

In Section D.1, we discuss why we find it better to model latent confounders at the donor level (via pseudobulking the gene expression vectors) rather than at the cell level. In Section D.2, we define the generator architecture used in sensGAN to compute the latent confounder for each donor.

D.1. Rationale for modeling scRNA-seq data on a donor level

We explain why we model sensGAN at the “donor level” (pseudobulking all cells from each of the n donors, despite having scRNA-seq data) rather than at the “cell level.” Specifically, this means we sum all the gene expression vectors of each cell to represent the gene expression vector for donor i , for $i \in \{1, \dots, n\}$. Our choice to pseudobulk the data might seem counterintuitive, since there are existing methods such as causarray (Du et al., 2026) and CoCoA-diff (Park & Kellis, 2021) that model latent confounders at the cell level. However, we note some important distinctions that motivate our choice of pseudobulking the data, despite having scRNA-seq data:

- **Intended goal of adjusting for latent confounders that represent future neuropathologies:** As we mention in Section 1.1 in the main text, one of the primary motivations of sensGAN is to have the estimated latent confounders represent potential co-occurring neurodegenerative diseases. This is fundamentally a donor-level quantity. In contrast, methods such as causarray (Du et al., 2026) and CoCoA-diff (Park & Kellis, 2021) learn cell-level confounders, which do not scientifically represent the same goal as our intended goal.
- **Considerations for the complexity of sensGAN architecture and computational efficiency:** Consider the architecture of the generator, which we describe in Section D.2. If we model the data at the cell level, we would need to adjust the architecture so that the donor-level confounder can be “broadcasted” to all cells from that donor. While this is feasible, it would dramatically increase the computational burden of fitting sensGAN and, hence, be detrimental to the goal of learning latent confounders.

D.2. Generator architecture

The sigmoid-head generator directly maps the input to a bounded confounder via a deterministic neural network:

$$\begin{aligned} h &= \phi(W_1[X, D, Y] + b_1), \\ \tilde{Z} &= \sigma(W_2h + b_2), \end{aligned}$$

where $\phi(\cdot)$ denotes a ReLU activation and $\sigma(\cdot)$ is the elementwise sigmoid function. This parameterization produces $\tilde{Z}$ deterministically and is simple and stable to optimize. In practice, we initialize the final-layer weights (W_2, b_2) by fitting a ridge regression in logit space, so that $\sigma(W_2h + b_2) \approx Z^{(0)}$, where $Z^{(0)}$ is an initial confounder estimate.

E. Loss details

Here, we include more details about the explicit calculations of the loss terms for the different components of sensGAN. In Section E.1, we define the likelihood loss used in the treatment predictor and the outcome predictor. In Section E.2, we define all the terms of the regularization term $\mathcal{L}_{\text{reg}}$ used when training in the generator in the adversarial training step of sensGAN.

E.1. Likelihood Loss

For the treatment predictor $f_D(\cdot)$, we use the binary cross-entropy loss with logits. Let $\hat{s}_i \in \mathbb{R}$ denote the predicted logit for donor i , and let $\hat{\pi}_i = \sigma(\hat{s}_i)$ be the corresponding predicted treatment probability, where $\sigma(\cdot)$ denotes the logistic function. The per-donor loss is given by

$$\ell_i^{\text{BCE}} = \max(\hat{s}_i, 0) - \hat{s}_i D_i + \log(1 + e^{-|\hat{s}_i|}),$$

which is a numerically stable implementation of binary cross-entropy with logits loss.

For the outcome predictor $f_Y(\cdot)$, we model gene expression counts with a negative binomial distribution. Let Y_{ij} denote the count for donor i and gene j , with mean $\mu_{ij} > 0$ and dispersion $\alpha_g > 0$. Define $r_j = 1/\alpha_j$. The per-entry negative log-likelihood is

$$\ell_{ij}^{\text{NB}} = - \left[\log \Gamma(Y_{ij} + r_j) - \log \Gamma(r_j) - \log \Gamma(Y_{ij} + 1) + r_j \log \left(\frac{r_j}{r_j + \mu_{ij}} \right) + Y_{ij} \log \left(\frac{\mu_{ij}}{r_j + \mu_{ij}} \right) \right].$$

We aggregate across genes to obtain a per-donor loss,

$$\ell_i^{\text{NB}} = \sum_{j=1}^p \ell_{ij}^{\text{NB}},$$

When training the most powerful predictor, the loss function used to train both $f_D(\cdot)$ and $f_Y(\cdot)$ is,

$$\mathcal{L}_L = \lambda_{\text{BCE}} \cdot \mathcal{L}_{\text{BCE}} + \lambda_{\text{NB}} \cdot \mathcal{L}_{\text{NB}}, \quad \text{where} \quad \mathcal{L}_{\text{BCE}} = \frac{1}{n} \sum_{i=1}^n \ell_i^{\text{BCE}}, \quad \mathcal{L}_{\text{NB}} = \frac{1}{n} \sum_{i=1}^n \ell_i^{\text{NB}}.$$

Table 2. Pearson correlation between estimated latent variable $\hat{Z}$ and simulated Z across κ targets under different regularization settings.

κ Target	Full Reg.	No Var.-Matching Reg. (4)
0.0	0.401	0.391
0.1	0.503	0.357
0.2	0.566	0.543
0.3	0.625	0.727
0.4	0.692	0.701
0.5	0.690	0.676
0.6	0.755	0.802
0.7	0.755	0.602
0.8	0.790	0.646
0.9	0.789	0.648
1.0	0.770	0.699

E.2. Regularization term

When training sensGAN via alternating adversarial optimization, we include a regularization term $\mathcal{L}_{\text{reg}}$ to stabilize generator updates and prevent degenerate confounder solutions. Let $\tilde{Z} \in \mathbb{R}^{n \times k}$ denote the current generator output and $\hat{Z} \in \mathbb{R}^{n \times k}$ the strongest confounder used as a reference. The regularization consists of the following components.

The correlation regularizer

$$\mathcal{L}_{\text{corr}}(\tilde{Z}, \hat{Z}) = \frac{1}{k} \sum_{a=1}^k (1 - \text{Corr}(\tilde{Z}_{\cdot a}, \hat{Z}_{\cdot a}))^2 + \frac{1}{k(k-1)} \sum_{a \neq b} \text{Corr}(\tilde{Z}_{\cdot a}, \hat{Z}_{\cdot b})^2,$$

encourages each generated confounder dimension to align with its corresponding reference dimension while discouraging cross-dimensional mixing.

The cosine similarity regularizer

$$\mathcal{L}_{\text{cos}}(\tilde{Z}, \hat{Z}) = 1 - \frac{\langle \text{vec}(\tilde{Z}), \text{vec}(\hat{Z}) \rangle}{\|\text{vec}(\tilde{Z})\|_2 \|\text{vec}(\hat{Z})\|_2},$$

promotes global alignment between the generated and reference confounders up to scale.

Finally, the variance-matching regularizer

$$\mathcal{L}_{\text{var}}(\tilde{Z}, \hat{Z}) = \frac{1}{k} \sum_{a=1}^k \left(\text{Var}_n(\tilde{Z}_{\cdot a}) - \text{Var}_n(\hat{Z}_{\cdot a}) \right)^2 \quad (4)$$

encourages the generated confounders to match the marginal scale of the reference confounder while preventing $\tilde{Z}$ from collapsing to trivial or near-constant values.

We combine the regularizers into a single term,

$$\mathcal{L}_{\text{reg}} = \lambda_{\text{corr}} \cdot \mathcal{L}_{\text{corr}} + \lambda_{\text{cos}} \cdot \mathcal{L}_{\text{cos}} + \lambda_{\text{var}} \cdot \mathcal{L}_{\text{var}},$$

where $\lambda_{\text{corr}}, \lambda_{\text{cos}}, \lambda_{\text{var}} \geq 0$ are tuning parameters.

We include an ablation study demonstrating that regularization stabilizes training. Removing variance-matching regularization leads to less consistent recovery of latent variables across κ levels (Table 2), whereas full regularization yields smoother behavior. Importantly, the resulting sensitivity contours and gene categorizations remain largely unchanged, indicating that regularization primarily stabilizes optimization and latent variable estimation rather than altering the final scientific conclusions.

F. Hyperparameters and training details

In Section F.1, we document all the hyperparameters used in sensGAN. In Section F.2, we describe in detail any additional training details of sensGAN not mentioned in the main text. In Section F.4, we document the various ways we find empirically beneficial when post-processing the results after Step 2 of sensGAN (i.e., adversarially training the predictors and generator).

F.1. Hyperparameters for loss functions

Optimization and training schedule. sensGAN alternates between refitting two predictive models and updating a generator. All components are optimized using the Adam optimizer as implemented in PyTorch, using learning rates specified below; we use `zero_grad()` only to clear accumulated gradients between backpropagation steps.

Predictor refitting. At each outer iteration, we refit the treatment predictor $f_D(\cdot)$ and the outcome predictor $f_Y(\cdot)$ on the current generated confounder $\tilde{Z}$. For both predictors, `lr` denotes the optimizer learning rate, `epochs` the maximum number of training epochs, and `tol` a convergence tolerance used to stop training early when improvements fall below a threshold.

- **Treatment predictor $f_D(\cdot)$.** We train f_D with the binary cross-entropy loss with logits using a learning rate `lr_cls` = 2×10^{-1} and at most `epochs_cls` = 10^4 epochs. We use a validation split `split_cls` = 0.7 and early stopping with patience `patience_cls` = 200 and tolerance `tol_cls` = 10^{-3} .
- **Outcome predictor $f_Y(\cdot)$.** We train f_Y by maximizing the negative binomial likelihood with a learning rate `lr_nb` = 2×10^{-1} and `epochs_nb` = 10^3 epochs.

Generator updates. The generator $g(\cdot)$ is trained in an alternating fashion with the predictors. Training proceeds over `gen_steps` = 10 outer epochs. Within each outer epoch, the predictors are first refit on the current $\tilde{Z}$, after which the generator is updated for up to `epochs` = 100 gradient steps using Adam with learning rate `lr_gen` = 10^{-3} .

Generator loss weights. We distinguish hyperparameters used for *predictor fitting* (above) from those used for the *generator objective*. In adversarial training, we penalize deviations from target predictive gains for D and Y with weights $\lambda_D = 1$ and $\lambda_Y = 1$, and apply scaling factors $s_D = s_Y = 10$ to normalize the magnitude of predictive-gain residuals. We additionally regularize the generator output to encourage structural alignment with the reference confounder via $\mathcal{L}_{\text{corr}}$, $\mathcal{L}_{\text{cos}}$, $\mathcal{L}_{\text{var}}$ with weights $\lambda_{\text{corr}} = \lambda_{\text{cos}} = \lambda_{\text{var}} = 10^{-3}$. Unless otherwise stated, these hyperparameters are fixed across experiments and chosen for optimization stability rather than performance tuning.

F.2. Additional training details

Initialization and warm start. We initialize the generator by fitting $g(\cdot)$ to reproduce the most powerful confounder $\hat{Z}$. We then warm-start the treatment and outcome predictors once using $\hat{Z}$ before entering the outer training loop. No separate warmup stage is used beyond this initialization.

Best-snapshot selection. In practice, the alternating optimization may fail to simultaneously achieve the target predictive gains for both D and Y within a finite number of outer epochs. To ensure a stable and informative output in such cases, we track the generator state that attains The lowest total generator loss throughout training. Specifically, at each generator update, we record the current $\tilde{Z}$ and associated statistics whenever the total loss improves upon the previously observed minimum. If the final training epochs do not meet the prescribed DBPR targets, we return the best-performing snapshot rather than the last iterate. This strategy prevents unstable late-stage updates from degrading the inferred confounder and provides a conservative estimate of the strongest confounder identified during training.

F.3. Bilevel optimization as an alternative training strategy

The current sensGAN training procedure follows a minimax-style adversarial optimization, where the generator and predictors are updated in an alternating fashion. While this formulation is natural for adversarial learning, minimax optimization could exhibit training/numerical instability. In particular, alternating updates approximate a simultaneous game,

where each component reacts only to the current state of the others, potentially leading to cyclical updates during training rather than a stable convergence.

An alternative formulation is to cast sensGAN as a bilevel optimization problem. In this setting, the predictors are treated as lower-level models that are optimized conditionally on the current latent confounder estimate, while the generator acts as the upper-level optimizer:

$$\min_G \mathcal{L}_{\text{sens}}(G, \theta_D^*(G), \theta_Y^*(G))$$

subject to

$$\theta_D^*(G), \theta_Y^*(G) = \arg \min_{\theta_D, \theta_Y} \mathcal{L}_{\text{pred}}(G, \theta_D, \theta_Y),$$

where θ_D and θ_Y denote the treatment and outcome predictors, respectively. Conceptually, the generator no longer reacts to partially optimized predictors at each iteration. Instead, it updates while explicitly accounting for the predictors' optimal response to the current latent variable estimate. This corresponds to a hierarchical "leader-follower" formulation rather than a simultaneous adversarial game. From a coding implementation point of view, this requires sensGAN to differentiate through the lower-level optimization through unrolled optimization scheme when updating the parameters of the generator. We believe this formulation provides a more stable training strategy for sensGAN. Since the generator objective depends on predictive gain constraints, unstable predictor updates can propagate directly into the latent variable estimation procedure, leading to oscillatory behavior. Importantly however, this changes the optimization procedure rather than the scientific objective itself.

We thank the ICML reviewers for suggesting this direction. While bilevel optimization is not explored in the current manuscript, we view it as a promising extension for improving optimization stability and reproducibility in future versions of sensGAN.

F.4. Post-processing of sensGAN results

Calibration via leave-one-out predictive gains. measured covariate. For each covariate X_i , we fit a leave-one-out (LOO) unadjusted NB GLM that excludes X_i , i.e. $(X_{-i}, D) \rightarrow Y$, and compare it to the corresponding unadjusted model that includes all measured covariates. The normalized predictive gain for outcome Y , i.e. η_{LOO} , is calculated based on these models, which captures the marginal predictive gain attributable to X_i . Repeating this procedure for all measured covariates yields an empirical distribution of η . We use this distribution as a calibration reference. Under the assumption that no latent confounder is stronger than a given measured covariate, we set the confounding strength parameter η (i.e., the confounding-effect knob) to match or remain below η_{LOO} .

Selection for contour visualization. For each target pair (η, κ) , sensGAN returns a generated confounder $\tilde{Z}$ and the corresponding achieved predictive gains (η', κ') . Because alternating optimization may not exactly hit the prescribed targets, we apply a feasibility filter before constructing contour plots: we retain a run only if both gains are within a fixed tolerance,

$$|\eta' - \eta| \leq \epsilon \quad \text{and} \quad |\kappa' - \kappa| \leq \epsilon, \quad \text{where } \epsilon = 0.1.$$

Only feasible runs are included when aggregating statistics (e.g., significance counts or p-value summaries) over the (η, κ) grid; targets that fail this criterion are treated as missing and are not plotted.

Isotonic smoothing. To summarize trends as a function of predictive gain, we additionally apply an isotonic regression transform to obtain a monotone estimate of the relationship between $-\log_{10} p$ and the achieved treatment gain κ' . Concretely, we fit a non-increasing isotonic regression model with clipping for out-of-range inputs,

$$\hat{m}(\cdot) = \text{IsoReg}(\text{decreasing}, \text{out_of_bounds} = \text{clip}).$$

Then, we report $\hat{m}(\kappa')$ as the calibrated curve. This monotone post-processing reduces local noise across grid evaluations while preserving the expected directional effect.

Refitting p-values for real data analyses. For real-data applications (SLE in Appendix Section J and AD in Section 3.3), we refit the outcome models (`R:glmGamPoi`) using the inferred confounder $\tilde{Z}$ to obtain final statistical significance measures. The refitting step yields calibrated p-values that account for overdispersion and mean-variance relationships in count data while decoupling statistical inference from the neural network training procedure.

Algorithm 1 sensGAN: adversarial sensitivity analysis with predictive-gain constraints

Input: disease status $D \in \{0, 1\}^n$, covariates $X \in \mathbb{R}^{n \times d}$, pseudobulk counts $Y \in \mathbb{N}^{n \times p}$, confounder dimension k , target $\eta^* \in [0, 1]$, sequence $\mathcal{K} \subset [0, 1]$, significance level δ , training tolerance ϵ .

Output: worst-case confounders $\{\tilde{Z}(\kappa, \eta)\}$, worst-case p-values $\{\tilde{P}(\kappa, \eta)\}$, and sensitivity contours.

Step 1A: Fit unadjusted models.

Fit a GLM regressing D on X to obtain $\hat{\pi}_{\text{un}}$.

Fit GLMs regressing Y on (X, D) to obtain $\hat{\mu}_{\text{un}}$, $\hat{\alpha}_{\text{un}}$, and $\text{SE}(\hat{T}_{Y, \text{un}})$.

Step 1B: Learn the most powerful confounder $\hat{Z}$.

Initialize treatment predictor $f_D(\cdot) : \{X, Z\} \mapsto D$.

Initialize outcome predictor $f_Y(\cdot) : \{X, D, Z\} \mapsto Y$.

Initialize generator $g(\cdot) : \{X, D, Y\} \mapsto \hat{Z} \in (0, 1)^{n \times k}$.

repeat

Train $f_D(\cdot)$ with current $\hat{Z}$ to minimize $\mathcal{L}_{\text{BCE}}$.

Train $f_Y(\cdot)$ with current $\hat{Z}$ to minimize $\mathcal{L}_{\text{NB}}$, using $\hat{\alpha}_{\text{un}}$.

Train $g(\cdot)$ to minimize $\mathcal{L}_L = \lambda_{\text{BCE}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{NB}}\mathcal{L}_{\text{NB}}$.

until convergence

Compute maximal DBPRs $(\hat{R}_D^2, \hat{R}_Y^2)$ using $\hat{Z}$.

Step 2: Learn worst-case confounders under predictive-gain constraints.

for fixed η^* and each $\kappa \in \mathcal{K}$ **do**

repeat

Generate current confounder $\tilde{Z} = g(X, D, Y)$.

Compute normalized predictive gains $\kappa' = \hat{R}_D^2 / \tilde{R}_D^2$ and $\eta' = \hat{R}_Y^2 / \tilde{R}_Y^2$.

Train $f_D(\cdot)$ with current $\tilde{Z}$ to minimize $\mathcal{L}_{\text{BCE}}$.

Train $f_Y(\cdot)$ with current $\tilde{Z}$ to minimize $\mathcal{L}_{\text{NB}}$, using $\hat{\alpha}_{\text{un}}$.

Train $g(\cdot)$ to minimize $\mathcal{L}_S + \mathcal{L}_P + \mathcal{L}_{\text{reg}}$ using δ , $\text{SE}(\hat{T}_{Y, \text{un}})$, $(\hat{R}_D^2, \hat{R}_Y^2)$, and targets (κ, η^*) .

until $|\kappa' - \kappa| \leq \epsilon$ and $|\eta' - \eta^*| \leq \epsilon$

Save $\tilde{Z}$ and the corresponding worst-case p-values for all p genes.

end for

G. sensGAN pseudocode

We present the pseudocode for sensGAN in Algorithm 1 to provide a high-level overview of its steps.

H. Data simulation details

In Sections H.1 and H.2, we describe how we simulate a binary and continuous-valued latent confounder Z , respectively. In Section H.3, we document our crawler, which is used to compute the reference sensitivity contours. In Section H.4, we document the results of how our crawler performs in our simulation setting.

H.1. Binary Z simulation

We first describe the simulation used to generate the results shown in Figures 4 and 4 and Table 1 in the main text, where we demonstrate that sensGAN is able to accurately estimate the latent confounders based on (κ^*, η^*) and that the genes are appropriately partitioned into different categories based on whether or not they are associated with the latent confounders. Here, we simulate the measured covariate matrix $X \in \mathbb{R}^{n \times d}$ and the latent binary confounder $Z \in \mathbb{R}^{n \times k}$ independently. Then, we simulate the treatment $D \in \mathbb{R}^{n \times 1}$ from a Bernoulli distribution with a logit link, while controlling for the variance contributed by terms involving X and Z . With the Negative Binomial GLM link function, we simulate the mean count matrix of scRNA-seq pseudocounts $Y \in \mathbb{R}^{n \times p}$. Then, we simulate the overdispersion rate to be a normal distribution centered around a constant vector $C \in \mathbb{R}^{p \times 1}$. Our simulated data can be compactly denoted as,

$$\begin{aligned} D &\sim \text{Bernoulli}(\sigma(XB_D + Z\Gamma_D)) \\ Y &\sim \text{NB}(\log(\mu), \alpha), \text{ where } \log(\mu) = DT_Y + XB_Y + Z\Gamma_Y, \end{aligned}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

With different combinations of $T_Y \in \mathbb{R}^{1 \times p}$ and $\Gamma_Y \in \mathbb{R}^{k \times p}$, we simulate 3 categories of genes with varied significance profiles:

- **Category i (“Significant”)**: Genes significantly associated with D but not with Z . Their expression remains significantly associated with D both with and without adjusting for Z . They are later referred to as the (always) significant genes.
- **Category ii (“Nullified”)**: Genes significantly associated with Z but not with D . These genes appear significant without adjusting for Z but become insignificant once Z is included in the model. They are later referred to as the nullified genes.
- **Category iii (“Insignificant”)**: Genes not significantly associated with either D or Z . They remain insignificant with and without adjustment for Z . They are later referred to as the (always) insignificant genes.

In this simulation, we set $n = 100$, $p = 100$, $d = 4$, $k = 1$, $C = 0.5$.

H.2. Continuous Z simulation for contour validation

Next, we describe the simulation used to generate Figure 5, which validates the contours of each gene. This simulation experiment validates the contour curves generated by sensGAN. Most simulation settings are similar to the above. The one difference is that we now simulate the continuous latent confounder $Z \in (0, 1)^{n \times k}$, where each entry is drawn independently and identically distributed from a uniform distribution. As mentioned in the main text, we use a continuous Z in this simulation to have finer control over predictive gains. In this experiment, we only simulate genes in **Category ii**, set $n = 100$, $p = 10$, $d = 4$, $k = 1$, $C = 0.5$. We use $p = 10$ so our contour validation works better empirically (see Appendix Section H.3 for details of our crawler) and so we can restrict attention to genes that are associated with Z but not with D . Although Category ii genes are simulated to be nullified by adjustment for Z , in practice, not all such genes are empirically nullified, since the outcome predictive gain of Z is constrained by the target η and finite-sample randomness can obscure marginal significance changes.

H.3. Crawler-based confounder generation.

To validate the contour generated by sensGAN, we implement a crawler-style worst-case search that directly explores the space of continuous confounders $Z \in [0, 1]^n$. This procedure is used to generate the *reference contours* shown in Figure 6 of the main text.

The crawler iteratively proposes candidate confounders using a diverse set of heuristics designed to cover a broad range of plausible confounding structures:

- **Independent random draws**: Sampling Z i.i.d. from $\text{Uniform}(0, 1)$.
- **Perturbations of the true confounder**: Adding noise to the simulated ground-truth Z to explore nearby confounding directions.
- **Rank-based transformations of D** : Constructing Z from the ranks of the treatment assignment, optionally with added noise.
- **Rank-based transformations of Y** : Using high-variance genes in Y to induce confounders correlated with outcome structure.
- **Principal components of (Y, D)** : Extracting leading principal components from the concatenated matrix of outcomes and treatment.
- **Jittered refinements of previous candidates**: Locally perturbing previously evaluated confounders to refine coverage of promising regions.

Each proposed confounder Z is evaluated by refitting adjusted models for D and Y and computing the corresponding deviance-based partial R^2 values. These are normalized to obtain predictive-gain coordinates (κ, η) . Candidates are then archived on a predefined (κ, η) grid, and for each grid cell, the crawler retains the confounder that minimizes the number of significant genes, yielding an explicit approximation to the worst-case confounding scenario.

H.4. Contour construction with crawler-based confounder.

To construct the reference contours, we discretize both κ and η into 11 equally spaced bins over $[0, 1]$, resulting in an 11×11 grid (Fig. 10). Only confounders whose achieved predictive gains fall within the corresponding bin are used to summarize that grid cell.

Table 3 reports the number of candidate confounders evaluated by the crawler that fall into each κ bin (± 0.05), illustrating coverage of the predictive-gain space.

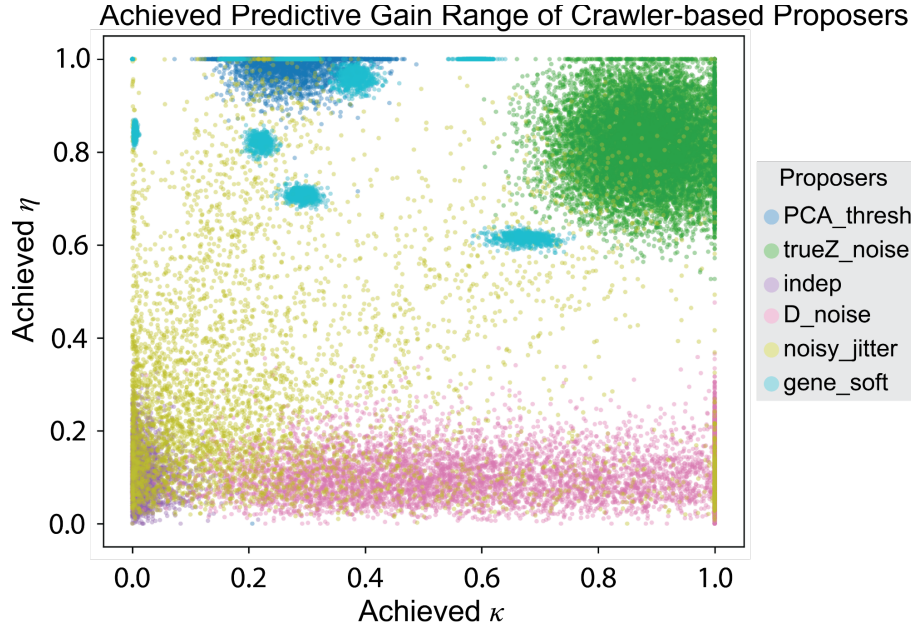


Figure 10. **Coverage of the (κ, η) predictive-gain space achieved by the crawler-based proposers.** Each point corresponds to a candidate confounder Z proposed by the crawler and evaluated by its normalized predictive gains for D (κ) and Y (η). Different proposal heuristics populate distinct regions of the grid, illustrating the heterogeneous and uneven coverage induced by direct search. This visualization motivates the need for sensGAN to more systematically and continuously explore the predictive-gain space.

Table 3. Number of crawler-generated confounders in each κ bin (± 0.05).

κ bin	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Count	42	72	60	37	21	30	32	27	20	5	8

I. Additional results: Simulation

Gene-level nullification contingency analysis. To evaluate sensGAN at the gene level by comparing per-gene sensitivity contour curves, we further summarize gene-level behavior with a contingency table that classifies genes as always significant, nullified, or always insignificant (Table 4). The contingency analysis shows that sensGAN accurately recovers the nullification status for the majority of genes, with consistent agreement across all three categories.

Effect size versus correlation structure. To further investigate the interpretation of sensitivity, we perform an additional simulation study examining the interplay between effect size and correlation structure. Building upon the simulation framework in Section 3.1, the genes here associated with the latent confounder (but not with the disease variable) are divided into two subgroups: (i) correlated genes with larger confounder effect sizes (mean ≈ 3 , std ≈ 2), and (ii) independent genes

Table 4. Gene-level nullification agreement between the crawler reference and sensGAN. The table reports the number of genes assigned to each significance category by sensGAN, stratified by the corresponding categories identified by the crawler-based reference. sensGAN correctly identifies 3 out of 4 genes that are nullified under worst-case confounding in the crawler analysis.

Method	Category	Significant	Nullified	Insignificant
sensGAN	Significant	1	0	0
	Nullified	0	3	1
	Insignificant	0	0	5

with smaller confounder effect sizes (mean ≈ 1.5 , std ≈ 2). We then compare their sensitivity contour behaviors under sensGAN. We observe that sensGAN preferentially nullifies correlated features, even when effect sizes are comparable. In particular, genes in subgroup (i) exhibit earlier sensitivity contour crossing points, becoming non-significant under smaller predictive gain requirements. These results suggest that sensGAN captures not only marginal effect size, but also the extent to which signals can be jointly explained by a shared latent structure. This behavior is consistent with many biological settings, where genes operate through coordinated pathways and shared regulatory programs.

J. Additional results: Systemic lupus erythematosus analysis

Systemic lupus erythematosus (SLE) is an autoimmune disease predominantly affecting women and individuals of Asian, African, and Hispanic descent (Perez et al., 2022). Here, the authors use multiplexed single-cell RNA sequencing to capture the complexity of immune cell populations and systematically profile the composition and transcriptional states of immune cells in a large multiethnic cohort. Our goal for this analysis is to demonstrate the rigor of sensGAN: withholding key donor covariates yields results similar to those from a differential expression analysis that uses them.

To remove genes with small variations, we preprocess the single-cell data by selecting the top 2,000 highly variable genes (HVGs) within each cell type, computing pseudobulk gene expression vector for each donor, and normalizing counts to the total library size per cell. With a focus on the T4 cell type, we aggregate single-cell expression profiles by summing counts across cells from the same subject, yielding a gene-level pseudobulk count matrix. Then, we remove genes with over 90% zero counts across subjects, retaining those expressed in at least 10% of donors for downstream analysis. To ensure method stability, we also remove genes whose counts do not converge in the naive GLM Negative Binomial Model. In total, the dataset includes $n = 256$ subjects (158 cases and 98 controls). For each subject, the case-control variable represents systemic lupus erythematosus (SLE) status, and the final number of retained genes is $p = 475$.

In the proof-of-principle analysis, we only provide the method with sex ($d = 1$) while holding out other measured covariates, including ethnicity and the sequencing batch information. To make sure the method captures biologically meaningful confounders, we decide to search for confounders ($k = 3$) with a wider range of confounding effects by letting κ and η be equal to each other and vary together, i.e., letting Z 's predictive gains on Y be equal to Z 's predictive gains on X . With a series of (κ, η) values from 0.2 to 1, with a step size of 0.05, the method identifies a series of confounders. We then perform adjusted differential expression analyses with confounders of varying strengths.

First, we analyze the significance profiles of genes under varying levels of confounder strengths. In the sensitivity contour plot (Fig. 11A), mean contour curves and 95% confidence intervals are colored by genes that are always significant, nullified, and always insignificant. Then, we calculate the exact κ at which the gene become insignificant by extrapolating on the stepwise contour curve (Fig. 11B). Shown by the bar graph of the number of differentially expressed genes in the κ s, we also observe that larger κ generally corresponds to fewer differentially expressed genes (Fig. 11C). These results show that the method can identify a series of confounders with varying strengths, and that more significant DEGs in the original model become insignificant as κ increases.

Then, we analyze the DEG sets across 3 methods to understand how sensGAN compares to existing methods while applying to real data. We use the Gamma-Poisson regression model to identify the DEGs associated with SLE, adjusting for different sets of covariates. First, for the sensGAN method, we adjust for sex and the sensGAN-learned confounders. Second, for the all-covariate method, we adjust for all measured covariates. Third, for the unadjusted method, we do not adjust for latent confounding but only adjust for sex. We present the results as a Venn diagram, in which circle areas correspond to the total number of DEGs identified while adjusting for different sets of covariates (Fig. 11D). The unadjusted DEG set is the largest, partially because it only adjusts for sex. The all-covariate DEG set is smaller than the unadjusted DEG set and larger than the sensGAN DEG set, partially due to its adjustment of other measured covariates. The sensGAN DEG set only includes the genes that are always significant with varying levels of confounding strengths; this is the smallest set and most of its

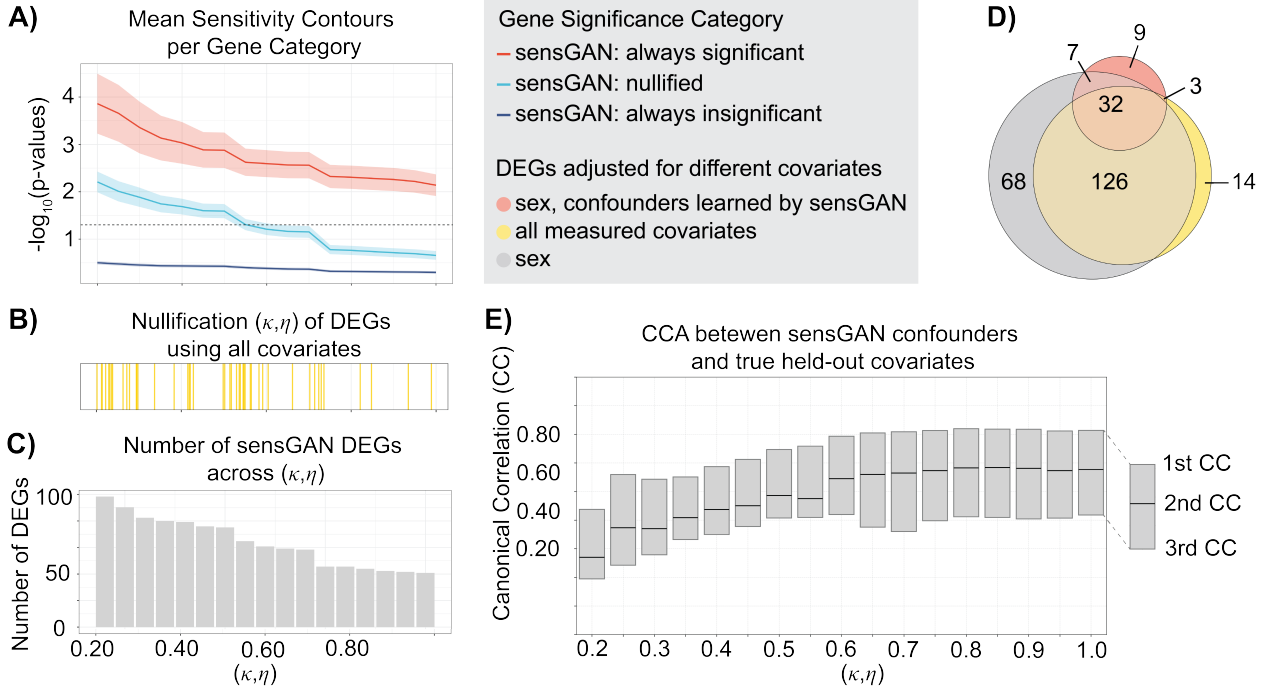


Figure 11. sensGAN identifies robust SLE DEGs under increasing latent confounding and learns biologically meaningful confounders aligned with held-out covariates. **A)** The sensitivity diagnostic contour plot with (κ, η) , where $\kappa = \eta$, varying from 0.2 to 1. **B)** The κ and η values where genes are nullified, extrapolated from the stepwise curve. **C)** The barplot of the number of DEGs at varying (κ, η) pairs. **D)** The Venn plot of significant DEGs calculated by the GLM Gamma-Poisson model adjusting for different covariates. **E)** The canonical correlations between the sensGAN confounders at varying levels of strength and the held-out covariates.

DEGs are also identified by the other methods.

Lastly, we summarize the relationships between the confounders ($k = 3$) learned by sensGAN and the held-out covariates ($d' = 4$) using canonical correlation analyses (CCA) (Fig. 11E). Every bar corresponds to 3 canonical correlation (CC) values while comparing the held-out covariates and the learned confounders. We observe that all CCs increase as the confounding strengths increase from 0.2 to 1. Based on this observation, we believe that the method can learn a series of biologically meaningful confounders that become increasingly similar to the held-out covariates as we allow stronger confounding effects.